

Gas Absorption in a Packed Bed

Final Lab Report
Unit Operations Lab 5
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1. Executive Summary (10 pts)

The objective of the lab is to characterize the fluid flow and mass transfer behavior of a packed bed gas absorber. Using a column with 1-cm Raschig rings, pressure drops were measured at different air and water flowrates. Additionally, carbon dioxide concentrations released from the sample valve were determined via sodium hydroxide titration to calculate mass transfer coefficients.

Experimental results yielded a void fraction of 0.82 at an operating temperature of 23.4 °C. Hydrodynamic analysis confirmed that the flooding point is highly sensitive to liquid load. At higher water flow rates, the system was more susceptible to pressure drops. As an example, for water flowrate at 8 L/min, flooding occurred at 10 L/min culminating in a pressure drop of 40.2 cm of red oil at an air flowrate of 20 L/min. In comparison, the dry system operated at the same air flow rate producing a negligible pressure drop of 0.2 cm. Through gas phase analysis via the Hempel apparatus, the volume fractions of carbon dioxide yielded 0.065 and 0.0675 for desorption and open loop absorption, respectively.

While open-loop absorption and desorption followed expected theoretical trends, the closed loop absorption exhibited anomalous behavior. Instead of increasing, the carbon dioxide concentration decreased from 0.0029 to a steady state value of 0.0014 gmol/L at 10 minutes. The most likely reason for this discrepancy stems from a second refilling of the reservoir and is reflected in the mass transfer coefficients, where the closed loop value (0.0382) was higher than the open loop's value (0.0156).

Overall, the results demonstrate that the hydrodynamic behavior of the packed bed absorber follows theoretical expectations, with pressure drop increasing with gas and liquid flow rates and clear transitions between operating states. However, some deviations were observed in the mass transfer data, particularly with the closed-loop absorption, where CO₂ concentration decreased instead of increasing as predicted in the theory. These discrepancies are likely due to experimental limitations such as titration sensitivity, flow instability, and sampling inconsistencies. Despite this, desorption and open loop results generally followed expected trends, highlighting the importance of accurate measurements and stable operating conditions in industrial gas absorption processes.

2. Introduction (10 pts)

Packed bed contactors are important equipment units in chemical engineering used in operations like gas absorption/stripping, catalytic reactions for high-conversion chemical reactions and distillation. The advantages of using these contactors are lower operating costs, they are efficient at higher temperatures and pressures and offer better mass transfer rates compared to other contactors [2]. These units are important in for removing toxic components from gas streams especially carbon dioxide to reduce greenhouse gas emissions. Additionally, distillation is an energy intensive process with distillation

responsible for approximately 40% of energy being used in the petrochemical industry [3].

The main objective of this experiment is to determine the characteristics of fluid flow, pressure drop, and flooding point in a packed bed gas absorber. Another goal is to determine the characteristics of gas absorption and overall mass transfer coefficient when removing carbon dioxide from an air stream including water in a packed bed gas absorber using 1-cm Raschig rings [1].

3. Theory

Gas absorption is a basic mass-transfer process where carbon dioxide as the solute is moved through the gas-liquid interface to water. A simple expression can be written for this: where N is the rate of transfer, k is the mass transfer coefficient multiplied with the unknown interfacial area and ΔC is the driving force of the concentration, mole fraction and partial pressure. [1].

To determine the hydraulic capacity of the column, the team calculated the column porosity which is the ratio of the void volume to the packed bed total volume [1].

The team also used two techniques to analyze the concentration of carbon dioxide absorbed. The first was the gas-phase analysis where the Hempl apparatus on the gas absorption panel was used volumetrically through comparing the initial sample volume to the remaining volume after the absorption in the 1.0 N NaOH solution. Additionally, the liquid-phase analysis where the titration with the 0.01N NaOH and phenolphthalein indicator shows if there is free carbon dioxide in the water sample and this is calculated with the formula:

An important objective of this experiment is to understand the characteristics of pressure drop under different flow patterns. For a dry bed with gas flowing across, the Ergun equation predicts the pressure drop per unit bed length (see below):

In this equation, μ is the gas viscosity, u is the superficial gas velocity, ϵ is bed porosity, D_p is the particle diameter, ρ is the gas density. The first term dominates at lower flow rates and velocities, while the second term dominates as the velocity increases and the effects of inertia begin [6].

When the water is introduced in a countercurrent flow, it takes up some void space reducing the area available for gas flow making the pressure drop higher than predicted with the Ergun equation. As the gas flow rate increases, the gas starts holding up liquid in the bed which leads to flooding and this displays as a steep break when plotting pressure drop vs. gas flow rate. The lab manual strongly recommends using 50% of the flooding velocity for the best contact between gas-liquid [1].

Henry's Law is needed to describe the equilibrium behavior of carbon dioxide between the gas and liquid phase:

In Henry's Law, p_{CO_2} is the partial pressure of CO_2 in the gas while C is the dissolved concentration in the water in (mol/l = M) and H represents Henry's constant (atm*L/mol). Henry's constant increases with temperature making it more important that the water temperature must be carefully recorded as it affects the equilibrium driving force also then affecting the mass transfer coefficients [6].

Furthermore, the Two Film Theory is used to measure the rate of CO_2 transfer. The assumption is that the interface is at equilibrium and the resistance to mass transfer is between two stagnant films on either side of the gas-liquid interface. The CO_2 flux through both films are expressed as [7].

Here, k_g and k_l are the mass transfer coefficients gas and liquid film, while the I represents the conditions at the interface. Henry's Law is then used to relate the interfacial concentrations and the two resistances which is combined to:

This shows the total resistance is the sum of the two film resistances in series. As CO_2 is absorbed in water, the Henry's constant is big enough where the liquid-film resistance controls making a bottleneck. What is reported is the overall volumetric coefficient where the a represents the interfacial area, because they both can't be separated unless there's further information [1].

In summary the patterns of behavior of CO_2 absorption during the transient state is governed by the magnitude of the concentration driving force. Initially, when the liquid contains little dissolved CO_2 , the driving force is large, resulting in rapid absorption. As the liquid approaches equilibrium, the driving force decreases and the rate of absorption slows, eventually reaching steady state.[4]

During desorption, the reverse process occurs, with CO_2 transferring from the liquid back into the gas phase. This process is similarly driven by a concentration gradient and exhibits a rapid initial rate followed by a gradual decline as equilibrium is approached.[5]

In an open-loop system, steady-state conditions are reached more quickly because fresh liquid continuously enters the column, maintaining a relatively constant driving force.[5]

4. Results (15 pts)

The purpose of this lab was to determine fluid flow, pressure drop, and the flooding point characteristics at different combinations of flowrates for air, carbon dioxide, and water in a packed bed absorber. With these characteristics known, including the type of bed

material (1-cm Raschig rings) and its void volume, next, gas absorption characteristics and the overall mass transfer coefficient for the removal of carbon dioxide from the bulk air into the bulk water can be determined.

All these characteristics are dependent on the interactions between the gas phase, liquid phase, and the 1-cm Raschig rings within the apparatus. Along with the guiding mass-transfer expression, , from [1]. The results from the experiment will be given below along with an explanation for what might be happening within the system based on calculated variables, graphs, and tabulated data.

Characterization of Liquid and Gas Flow in Packed Columns:

To understand the effect of the packed bed material in the column, the void volume (or fraction) was calculated using the equation from [1]. A final value of 0.82 was calculated indicating that this material provided adequate space for gas-liquid interaction which ideally would aid greatly in mass-transfer.

The values for this calculation are mentioned in Appendix section

To determine the pressure-drop behavior of the system, the apparatus was operated with different air and water flowrates that travel countercurrent to each other. The value of the pressure-drops was determined using a manometer filled with red oil that had a specific gravity of 8.26×10^{-1} . The pressure-drop values along with the visible behavior of the column also allowed for the determination of flooding within the column. Flooding is a very important parameter that tells the operator when the gas flowrate is too high compared to the liquid and it begins to forcefully lift itself and the liquid out of the bed. This leads to inadequate contact between the two phases and prevents the establishment of equilibrium mass-transfer. For the different combinations of air and water flowrate, different flooding points are seen which indicates that different flow regimes are occurring within the column.

The flooding point was determined based on the point where the slope of the plot increased the greatest. This point correlated with very turbulent behavior within the column during the experiment which shows a link between quantitative and qualitative data. On the plots, the flooding point is indicated with a blue star

The first trial was done with air flowing upward through the column only. With this set-up, the flooding point was reached at around 15.5 L/min of airflow based on a plot of the experimental data. The pressure drop at this flooding point was 140 cm of red oil.

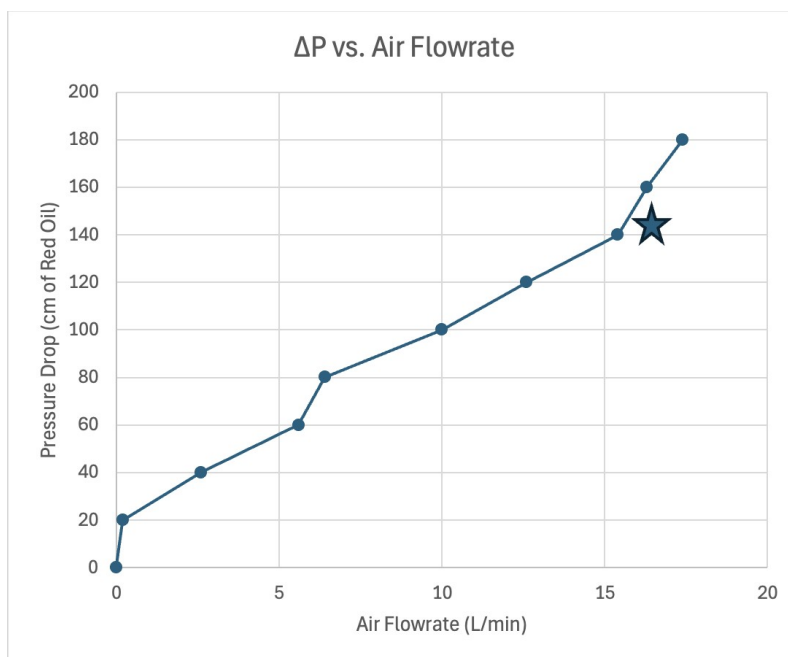


Figure 1. Pressure Drop through Packed-Bed with Airflow Only

The next trial involved different air flowrates flowing upward through the column while a constant volumetric flowrate of water at 1.5 L/min entered the system from the top of the column. For this flow regime, the flooding point was reached when ~80 L/min of airflow traveled through the packed bed. The pressure drop seen at this flooding point was ~18 cm of red oil.

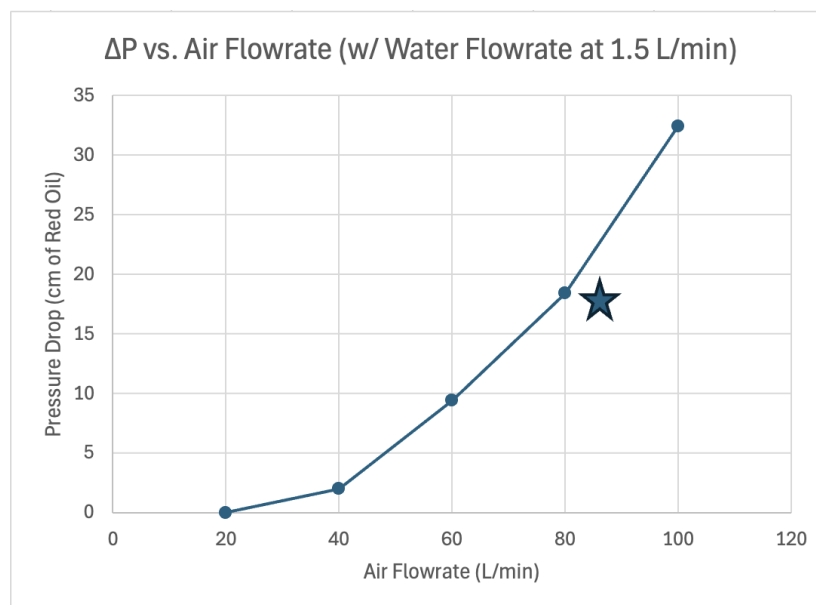


Figure 2. Pressure Drop through Packed-Bed with Waterflow at 1.5 L/min

The following trial's flow regime involved a constant volumetric flowrate of water at 3.0 L/min with airflow coming from the bottom. Airflow was steadily increased until the

flooding point was reached. For this trial, after analyzing the plot of the experimental data, it was concluded that the flooding point occurred when airflow was ~ 40 L/min. The pressure drop achieved at this flooding point was ~ 5 cm of red oil.

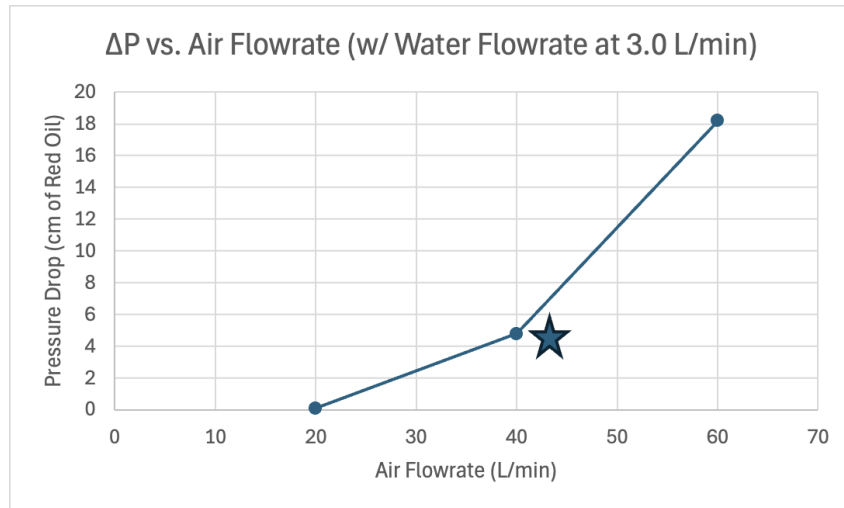


Figure 3. Pressure Drop through Packed-Bed with Waterflow at 3.0 L/min

Analyzing the next trial with a constant water flowrate of 6.0 L/min flowing from the top of the column with a steadily increasing air flowrate through the bottom, by looking at the plot of the experimental data, it was determined that the flooding point was reached when the airflow was ~ 20 L/min. The pressure drop seen at this flooding point was also ~ 5 cm of red oil.

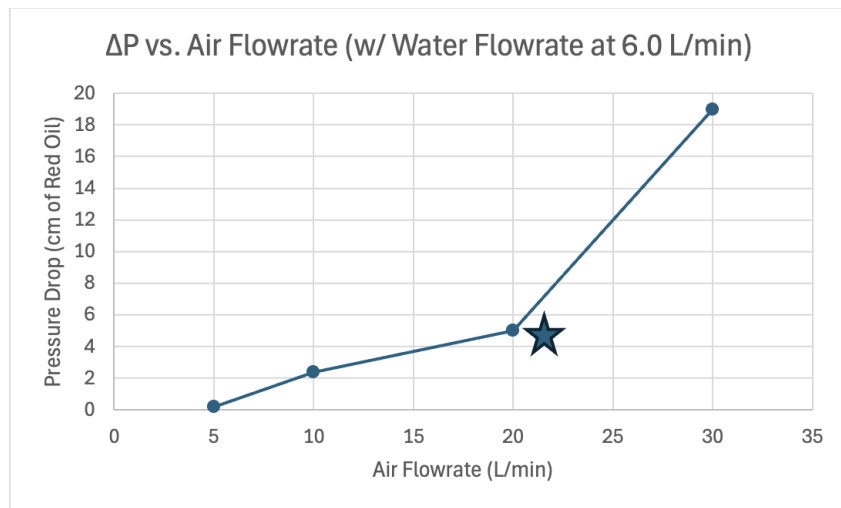


Figure 4. Pressure Drop through Packed-Bed with Waterflow at 6.0 L/min

The last flow regime analyzed involved a constant volumetric flowrate of water at 8.0 L/min. Through plotting the experimental data, it was determined that the flooding point was reached when the air flowrate was ~ 10 L/min. The pressure drop at this flooding point was ~ 12 cm of red oil.

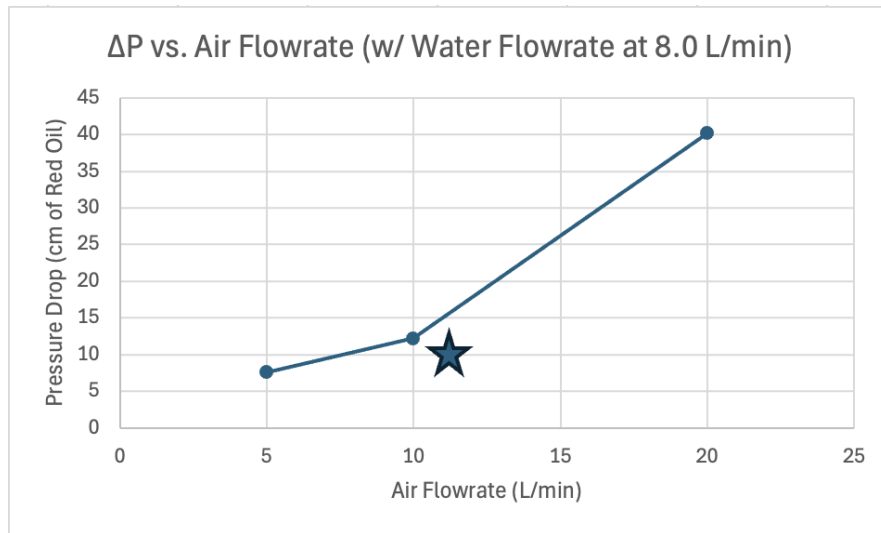


Figure 5. Pressure Drop through Packed-Bed with Waterflow at 8.0 L/min

Trends of the Different Flow Regimes:

After compiling the results of this characterization procedure into a table and then producing graphs of the flooding point values for the air flow and pressure drop vs. the water flow, it was noticed that there were noticeable trends. For the airflow flooding point values, initially the value rose up when water flow was introduced into the system. This initial jump sort of established a baseline for this parameter. But as the water flowrate began to increase, this value steadily decreased in an almost linear-like fashion. This observation hints that the flooding point air flowrate is inversely proportional to the water flowrate.

A similar trend was seen with the flooding point pressure-drop values when they were plotted against the water flowrate. Although, the decrease in values was not linear-like. Instead, the pressure-drop decreased but quickly plateaued and then rose for the last trail. Nevertheless, the data still showed some sort of inverse relationship between the two parameters as the water flowrate increased.

Table 1. Flooding Point Values vs. Water Flow

Trial	Water Flow (L/min)	Air Flow at F.P. (L/min)	ΔP at F.P. (cm of red oil)
1	0.0	15	140
2	1.5	80	18
3	3.0	40	5
4	6.0	20	5
5	8.0	10	12

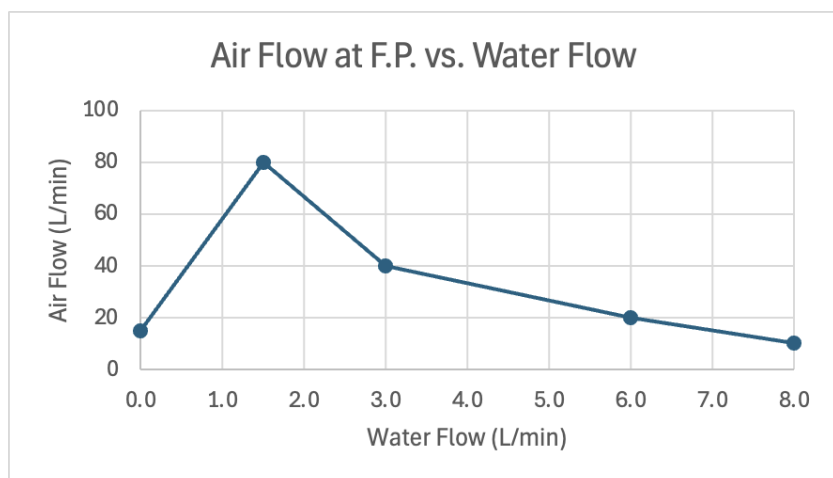


Figure 6. Flooding Point Air Flow vs. Water Flow

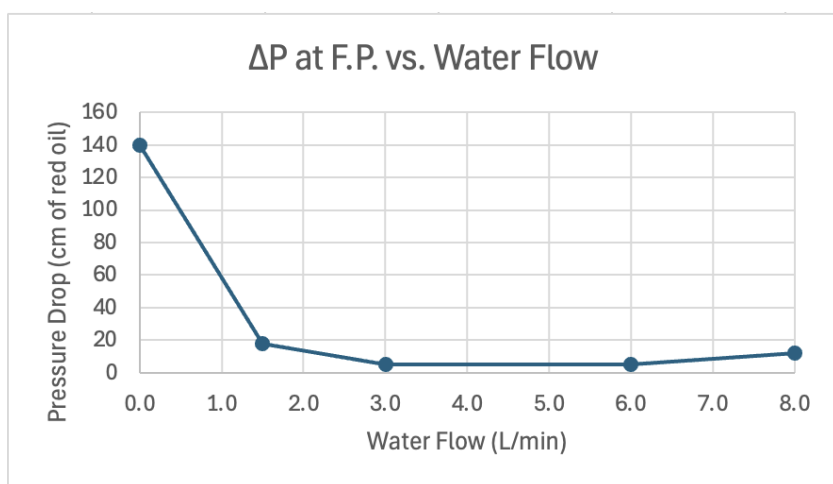


Figure 7. Flooding Point Pressure-Drop vs. Water Flow

Absorption and Desorption of Carbon Dioxide into Water:

For parts two and three of this lab, experiments were run to understand the absorption/desorption behavior of carbon dioxide into water.

Part two involved a water and air stream, mixed with carbon dioxide, being brought into equilibrium with each other in a closed-loop water system. Once equilibrium was achieved and adequate time was allowed for the mass-transfer of carbon dioxide into the water, next, desorption of carbon dioxide back into the air was studied. This was done by shutting off carbon dioxide flow and opening drain D1, near the water tank, and taking samples from S4.

Part three involved operating the apparatus to study the absorption of carbon dioxide by water in a continuous open-loop water system. This set-up was obtained by having water flow down the column while air (w/ carbon dioxide) is flowing upward. As the water

travels down the column and interacts with the gas allowing for mass-transfer, it is immediately drained out of D1 while samples are taken from S4.

To calculate the concentration of carbon dioxide in the water for both parts (and initially), the equation in appendix B from [1]] was used, , after 50 mL samples of water were titrated with 0.05 N NaOH (equivalent to 0.05 M NaOH) and a persistent pink color was attained. To determine the volume fraction of carbon dioxide leaving the column, the procedure in appendix A, from [1], for using the Hempl apparatus was followed. This volume fraction was then multiplied by the exiting molar flowrate of the air that was calculated using the ideal gas equation along with the incoming air flowrate and the measured pressure of the air read from the regulating valve. The temperature of the room was assumed to be 25°C for this calculation; it was also assumed that no accumulation of air occurred during operation. The initial concentration of carbon dioxide was calculated using the incoming flowrate, 4.0 L/min, the assumed room temperature, 25°C, and the ideal gas equation. All concentrations were in molar units (M)

Part Two Results:

For part 2a, the absorption of carbon dioxide by water in a closed-loop system was the focus. During the experiment, there was a partially open drain valve that was caught during the middle of the experiment. A substantial amount of water was lost from the tank and the apparatus. Fresh DI water had to be added to the tank during. Due to the chaos surrounding this part of the lab, the data gathered was faulty and showed that the concentration of carbon dioxide dropped before eventually reaching a steady state. This is the opposite of what would be expected in this situation due to the water having a smaller concentration of carbon dioxide compared to the air that's continuously being mixed with a constant stream of fresh carbon dioxide.

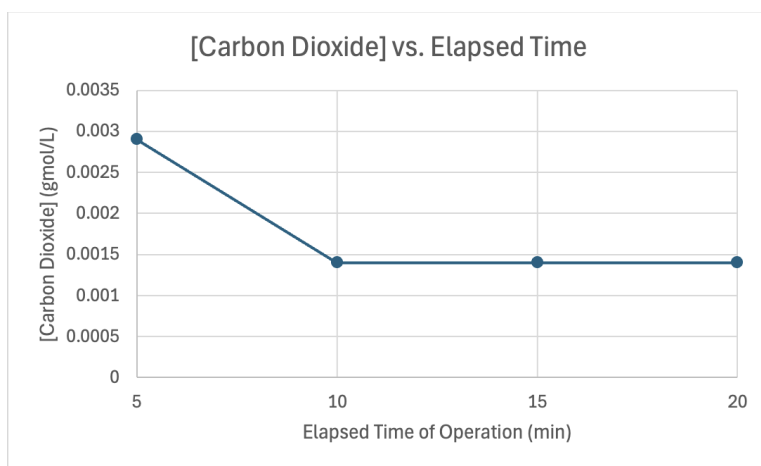


Figure 8. Carbon Dioxide Conc. vs Elapsed Time During Absorption (Closed-Loop)

Table 2. Absorption of Carbon Dioxide (Closed-Loop)

Absorption of CO ₂ (Closed Water Loop)			
Sample	NaOH Added (mL)	Cd (gmol of CO ₂ per L of free CO ₂)	Elapsed Time of Operation (min)
1	2.9	0.0029	5
2	1.4	0.0014	10
3	1.4	0.0014	15
4	1.4	0.0014	20

For part 2b, the desorption of carbon dioxide from the water was studied. As expected, the concentration of carbon dioxide dropped before reaching a steady state due to the source of carbon dioxide being cut off. This action leads to the development of there being a higher concentration of carbon dioxide in the water compared the surrounding gas. The driving force for mass-transfer is now reversed.

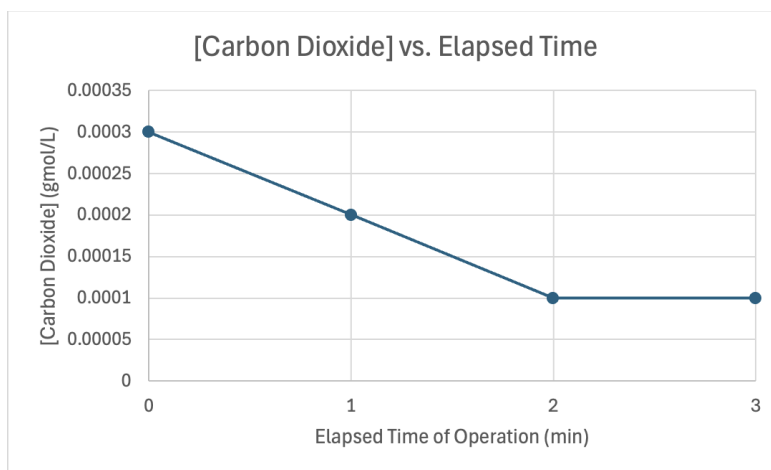


Figure 9. Carbon Dioxide Conc. vs Elapsed Time during Desorption (Closed-Loop)

Table 2. Desorption of Carbon Dioxide Over Time (Closed-Loop)

Desorption of CO ₂ from H ₂ O into Air			
Sample	NaOH Added (mL)	Cd (gmol of CO ₂ per L of free CO ₂)	Elapsed Time of Operation (min)
1	0.3	0.0003	0
2	0.2	0.0002	1
3	0.1	0.0001	2
4	0.1	0.0001	3

For part three of this lab, absorption of carbon dioxide in water was now being studied in an open-loop system. For this part of the lab, it's expected that, just as in part 2a, that carbon dioxide concentration will increase until a steady state is reached due to constant inflow of carbon dioxide into the apparatus creating a concentration gradient that favors mass-transfer into the water phase. The data for this part does show that, although, there are some fluctuations in values before a steady state is reached. Even though this is the case, the steady state value is higher than the initial reading value which shows that carbon dioxide concentration did increase overall.

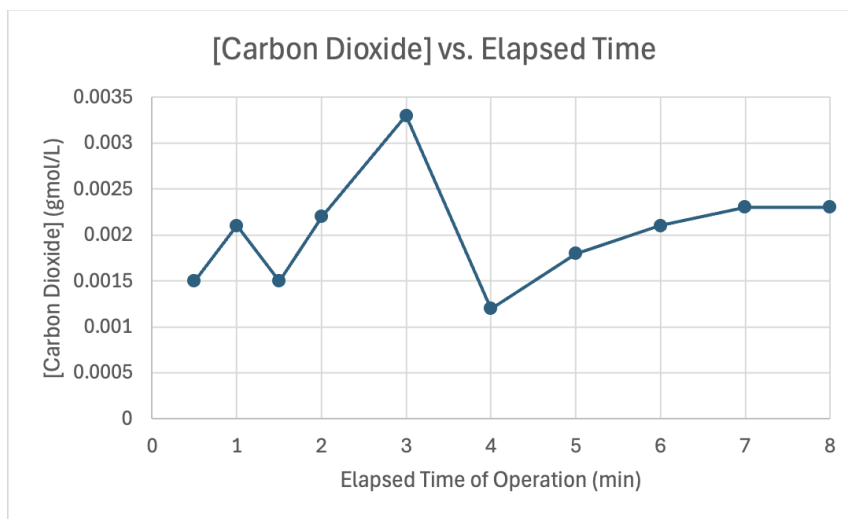


Figure 10. Carbon Dioxide Conc. vs Elapsed Time during Absorption (Open-Loop)

Table 3. Absorption of Carbon Dioxide during Absorption (Open-Loop)

Absorption of CO ₂ (Open Water Loop)			
Sample	NaOH Added (mL)	Cd (gmol of CO ₂ per L of free CO ₂)	Elapsed Time of Operation (min)
1	1.5	0.0015	0.5
2	2.1	0.0021	1
3	1.5	0.0015	1.5
4	2.2	0.0022	2
5	3.3	0.0033	3
6	1.2	0.0012	4
7	1.8	0.0018	5
8	2.1	0.0021	6
9	2.3	0.0023	7
10	2.3	0.0023	8

Overall Mass Transfer Coefficients:

Using the steady state data from the experiments, the overall mass transfer coefficient was calculated using the mass-transfer equation, , mentioned above from [1]. Technically there are two mass-transfer processes in this lab. There's mass transfer from the bulk gas to the gas/water interface and there's mass-transfer from the gas/water interface to the bulk water. Since this is the case, there's also two mass-transfer coefficients that can be calculated. Although, without knowledge of the carbon dioxide concentrations at the gas/water interface or the interfacial area, a , these parameters can't be calculated. So instead, the overall mass-transfer coefficient, K_a , which includes the interfacial area, is calculated. This parameter is calculated by focusing on mass-transfer from the bulk gas to the bulk water using the equation . Where mass transfer can be calculated using equation 1. or 2. . Both equations are from [8].

The sample calculations section in the appendix will demonstrate how the variables of these equations were calculated. The final values of are shown below. These values were calculated for the two portions of the lab where the absorption of carbon dioxide in water was studied, parts 2a and 3. Overall, it can be seen that absorption is favored when a closed-loop set-up is operated due to the constant recycling of the water throughout the whole apparatus allowing for multiple chances for the water and gas to interact while flowing through the packed-bed.

Table 4. Overall Mass-Transfer Coefficients

Mass Transfer Coefficients	
Mechanism	Ka (1/min)
Closed-Loop	0.0382
Open-Loop	0.0156

5. Discussion

The results experiment demonstrates how hydrodynamic and mass transfer behavior of a packed bed gas absorber. Overall, the pressure drop data follows theoretical behavioral outcomes listed, meanwhile the CO₂ absorption results exhibit deviations that can be explained by experimental limitations and human error interference.

The experimental results demonstrated that the pressure drop results closely follow theoretical expectations for packed bed systems. As shown in Figure 1, which presents pressure drop (ΔP) versus air flow rate without liquid flow, the relation between pressure drop increases steadily with different increasing gas flow rate, this is consistent with the Ergun equation. When liquid is introduced into the experiment, the trends shown in Figures 2,3,4, and 5 demonstrate progressively higher pressure drops and steeper slopes as liquid flow increases. The changes in the pressure drop that are related to the introduction of liquid also correspond to flooding of the system when pressure changes enough to overflow the apparatus.

At the low gas flow rates, the increase in pressure drop is slow and gradual, corresponding to the pre-loading theory and procedure. As gas flow increases, the slope becomes steeper, indicating the loading of the system. At the higher flow rates, particularly in Figures 4 and 5, the sharp increase in pressure drop suggests approach to the flooding condition. This behavior is consistent with theoretical packed bed hydrodynamics.

In contrast, the absorption results from the experiment do not fully agree with theoretical expectations. Such as in closed-loop absorption Figure , the CO₂ concentration decreases from approximately 0.0029 to 0.0014 gmol/L before leveling to a concentration, whereas

theory predicts an increasing trend over time as more CO_2 . This discrepancy indicates that the disagreement is likely due to experimental error rather than a failure of the theoretical behavior of absorption.

The desorption behavior shown in Figure 7 follows the expected theoretical trend, with CO_2 concentration decreasing over time, confirming that the system follows the expected mass transfer behavior under those conditions. The open-loop absorption data in Figure 8 shows fluctuating but generally increasing concentrations before stabilizing, partially aligning with theoretical expectations of rapid approach to steady state leading to the trends of overall rise and fluctuations.

The following discrepancies observed in the absorption data Figure 6 suggest limitations in the experimental setup and in the measurement techniques. The titration analysis method used to determine CO_2 concentration is highly sensitive to endpoint detection, and small inaccuracies in NaOH volume measurements can significantly affect calculated concentrations. These are usually the result of human error but in this case sampling inconsistencies also contributed. Since the liquid reservoir was not well mixed, the samples used for analysis may not represent the true bulk concentration. Additionally, the system had already approached equilibrium from the previous lab group prior to initial sampling, this could explain the observed decrease in concentration.

Gas flow instability is another likely source of error when flow rates were adjusted to meet the procedure requirements. Fluctuations in flow rates would directly impact both pressures drop such as in Figures 2,3,4, and 5 and mass transfer performance in Figures 6,7, and 8. Furthermore, non-ideal flow behavior such as channeling or incomplete wetting of the packing from error could have reduced effective interfacial area and limit absorption efficiency leading to the discrepancies in the data and the theory.

Improvements to the experimental design could include automated titration, computer composition and pH analysis, enhancing the mixing in the liquid reservoir, digital flow controllers for improved stability, and inline CO_2 sensors to reduce sampling error.

The analysis assumes steady-state operation conditions, but the fluctuations observed in the concentration data in Figures 6,7, and 8 suggest that this condition may not have been fully achieved. The assumption of uniform liquid distribution is unlikely to be true due to the apparatus design, as real packed beds often exhibit fixed channeling and maldistribution.

Additionally, the assumption of accurate concentration measurement is unclear due to the inconsistencies observed in Figure 6. Meanwhile, the assumption that liquid-phase resistance is the dominant force is theoretically reasonable, but the data fluctuations make it difficult to confirm experimentally.

Pressure drop data in Figures 1,2,3,4, and 5 provides strong evidence for the existence of distinct hydrodynamic behavior stages such as pre-loading, loading, and near-flooding conditions. The effect of increasing liquid flow rate on pressure drops and flooding behavior is clearly demonstrated with data relation in Figures 1,2,3,4, and 5.

However, conclusions regarding mass transfer behavior are more speculative due to the data results. While desorption trends Figure 7 align with theory, the inconsistency in absorption data such as in Figure 6 questions the calculated mass transfer coefficients. Similarly, to how the liquid-phase resistance dominates cannot be conclusively determined from the available data results.

From the broad perspective, the results confirm that operating conditions must be maintained below flooding limits, as indicated by the sharp increases in pressure drop observed in Figures 4 and 5. The relationship between experimental pressure drop trends and theory supports the hydrodynamic design correlations.

The discrepancies in mass transfer data demonstrate the importance of accurate measurement techniques in absorption systems. In industrial applications, reliable instrumentation and stable operating conditions are critical for achieving predictable performance. Such as the use of packed bed gas absorbers used in processes such as carbon capture from flue gases, natural gas sweetening, and removal of contaminants like SO_2 and NH_3 in chemical manufacturing. In these applications, operating conditions must be carefully controlled to maintain efficient mass transfer while avoiding flooding, which can reduce performance and increase operational risk.

The trends observed in this experiment, including the sensitivity of mass transfer to flow conditions and measurement accuracy, highlight the need for precise monitoring and optimization of gas and liquid flow rates, packing design, and process conditions in large-scale industrial systems. Theoretically the results support the behavior expectations for hydrodynamics and mass transfer while demonstrating the challenges of applying ideal assumptions to real systems.

Future work should investigate the relative contributions of gas phase and liquid phase resistances, the impact of packing characteristics on performance, and the effect of temperature on absorption behavior. Improvements in the experiment should be made in the control and increasing the number of trials, with these improvements this would help reduce uncertainty and provide more reliable data that would follow the thermotical expectations.

6. References

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7. Appendix I: Data Tabulation/Graphs*Table 1: Airflow with No Water*

Pressure Drops (cm of red oil)	Flowrate (L/min)
0	0
0.2	20
2.6	40
5.6	60
6.4	80
10	100
12.6	120
15.4	140
16.3	160
17.4	180

Table 2: Airflow with a Water Flowrate of 1.5 L/min

Pressure Drops (cm of red oil)	Flowrate (L/min)
0	20
2	40
9.4	60
18.4	80
32.4	100

Table 3: Airflow with a Water Flowrate of 3 L/min

Pressure Drops (cm of red oil)	Flowrate (L/min)
0.1	20
4.8	40
18.2	60

Table 4: Airflow with a Water Flowrate of 6 L/min

Pressure Drops (cm of red oil)	Flowrate (L/min)
0.2	5
2.4	10
5	20
19	30

Table 5: Airflow with a Water Flowrate of 8 L/min

Pressure Drops (cm of red oil)	Flowrate (L/min)
7.6	5
12.2	10
40.2	20

Table 6: Absorption of (Closed Water Loop)

Sample	NaOH Added (mL)	Cd (gmol of CO ₂ per L of free CO ₂)	Elapsed Time of Operation (min)
1	2.9	0.0029	5
2	1.4	0.0014	10
3	1.4	0.0014	15
4	1.4	0.0014	20

Table 7: Desorption of from into Air

Sample	NaOH Added (mL)	Cd (gmol of CO ₂ per L of free CO ₂)	Elapsed Time of Operation (min)
1	0.3	0.0003	0
2	0.2	0.0002	1
3	0.1	0.0001	2
4	0.1	0.0001	3

Table 8: Absorption of (Open Water Loop)

Sample	NaOH Added (mL)	Cd (gmol of CO ₂ per L of free CO ₂)	Elapsed Time of Operation (min)
1	1.5	0.0015	0.5
2	2.1	0.0021	1
3	1.5	0.0015	1.5
4	2.2	0.0022	2
5	3.3	0.0033	3
6	1.2	0.0012	4
7	1.8	0.0018	5
8	2.1	0.0021	6
9	2.3	0.0023	7
10	2.3	0.0023	8

Table 9: Volume Frac. Of in Air Leaving the Column

Part in Lab	NaOH (mL)	Air + CO ₂ (mL)	Volume Frac. Of CO ₂
2a	2.6	40	0.065
3	2.7	40	0.0675

Table 10: Mass Transfer

Mechanism	K _a (1/min)
Closed-Loop	0.0382
Open-Loop	0.0156

Table 11: Flooding Point and Pressure Drops

Trial	Water Flow (L/min)	Air Flow at F.P. (L/min)	ΔP at F.P. (cm of red oil)
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1	0.0	15	140
2	1.5	80	18
3	3.0	40	5
4	6.0	20	5
5	8.0	10	12

Table 12: Parameters and Void Fraction

Temperature of Water (C)	23.4
Compressed Air Pressure (psig)	42
Void Fraction	0.82

8. Appendix II: Error Analysis

The main source of quantitative errors come from limitations related to measurements. Titration readings had a ± 0.05 mL NaOH error, having a negative effect at low concentrations. The 0.1 mL readings in the closed-loop desorption data has a $\pm 50\%$ uncertainty which means the 0.001 gmol/L reporting could vary from 0.00005 to 0.00015 gmol/L. However, the larger measurements like 3.3 mL open loop only have a $\pm 1.5\%$ uncertainty. Another error was through the pressure drop readings from the manometer having a 0.1 cm resolution resulting in a 100% relative uncertainty when reading at low flow near zero, however, only 0.25% at the 40.2cm reading at 8 L/min water. The flooding point contains an uncertainty of at least ± 10 L/min because it was observed between different data points instead of a continuous reading. Furthermore, the Hempl apparatus introduced an estimated $\pm 2\%$ relative uncertainty on the carbon dioxide volume fraction readings of 0.065 and 0.075 from parts 2a and 3, and this is due to the ± 0.5 mL syringe resolution over a 40 mL sample. Overall, the errors accumulated over time spreading to the mass transfer coefficients which showed a 60% difference between the open (= 0.0156 min^{-1}) and closed loop (= 0.0382 min^{-1}) systems which is a large gap.

The major source of experimental error was the reservoir refill during our first run causing the concentration to decrease which is observed in the open-loop data. The concentration dropped from 0.033 gmol/L at sample 5 to 0.0012 gmol/L at sample 6 which is unexpected of absorption behavior. The new deionized (DI) water entering the system had minimal (near-zero) dissolved CO_2 and already diluted the accumulated CO_2 in the loop. This then favors the open loop calculations for the mass transfer coefficients. Upon further trials, the team noticed that this error can be avoided by filling the reservoir before starting and sizing the run to complete in the available water volume. After this, the CO_2 degassing from collected samples before any titration may have caused for the lack of reporting of the dissolved concentrations as the CO_2 starts to escape when a sample is exposed to atmospheric pressure and this is seen with the drop from 0.0003 to

0.0001gmol/L in 3 minutes of desorption data. Additionally, the titration endpoint also introduced uncertainty as the light pink color change is left to visual understanding especially at the NaOH volumes 0.1 to 0.2 mL in which a single extra drop affects results by 30-50%. Lastly, underreported air flow drift altered the superficial gas velocity and the CO₂ partial pressure driving force in between the readings affects the mass transfer coefficient calculations and pressure drop curves.

9. Appendix III: Sample Calculations

Calculating the Conc. of Carbon Dioxide in Water (Sample 1; Closed-Loop):

= 2.9 mL
 = 0.05 N (same as 0.05 M)
 = 50 mL

Calculating the Initial Conc. of Air Flowing into the system:

=>

P = 42 psig = 3.86 atm
 = 20
 T = 25°C = 298 K
 R = 0.08206

Multiplying this value by the volume fraction of carbon dioxide determined from the Hempl apparatus roughly gives the concentration of carbon dioxide leaving the apparatus

The same process is done for calculating the concentration of carbon dioxide entering the system

Calculating the Overall Mass Transfer Coefficient (Closed-Loop):

= 6.0
 = 0.0014 **from **
 = 0.0001 **calculated using the equation for the (0.1 mL) needed to titrate the initial amount of carbon dioxide in the water**
 = 0.065 (volume fraction of carbon dioxide from Hempl apparatus) x (for incoming air; calculated above) = 0.2054
 = 0.0014
 = 0.0078

Calculation of Void Fraction for Bed Material:

= 369 mL **volume of the bed material**
 = 450 mL **volume required to fill void space of bed material**

10. Appendix IV: Individual Team Contributions

	TIME (HOURS)	DESCRIPTION
Operator (Both Lab Days)	6 hours	Joshua Laforest, Cesearo Dilosa, Javier Silva, Ify Bidokwu
Pre-Lab Editing	40 minutes	Joshua Laforest, Cesearo Dilosa, Javier Silva, Ify Bidokwu
Experimental Objective (Prelab)	10 minutes	Joshua Laforest
Apparatus (prelab)	1 hour	Joshua Laforest, Javier Silva
Materials & Supplies (prelab)	40 minutes	Javier Silva
Procedure (prelab)	1.5 hours	Cesearo Dilosa
Project Safety Evaluation (prelab)	1 hour	Ify Bidokwu
Final Lab Editing	1 hour	Cesearo Dilosa, Javier Silve, Josh Laforest, Ify Bidokwu
Executive Summary (final lab)	40 minutes	Joshua Laforest
Introduction (Final lab)	45 minutes	Ify Bidokwu
Theory (Final Lab)	2 hours	Ify Bidokwu, Javier Silva
Results (final lab)	3.5 hours	Cesearo Dilosa
Discussion (final lab)	2 hours	Javier Silva
References (final and/or prelab)	20 minutes	Ify Bidokwu, Javier Silva, Cesearo Dilosa
Data Tabulation/Graphs (final)	50 minutes	Joshua Laforest, Cesearo Dilosa
Error Analysis (final lab)	45 minutes	Ify Bidokwu
Sample Calculations (final Lab)	30 minutes	Cesearo Dilosa
Power Point Presentation	45 minutes	Joshua Laforest, Cesearo Dilosa, Javier Silva, Ify Bidokwu
Total	24 hours	